Transforms-Based Bayesian Tensor Completion Method for Network Traffic Measurement Data Recovery

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***Abstract*—Network traffic measurement is regarded as the bedrock of next-generation network systems. Its purpose is to monitor the network traffic and provide data support for traffic engineering. For this reason, monitoring traffic data from a network-wide perspective is particularly important. However, the proliferation of network services has led to the explosive growth of network traffic, which has brought significant challenges to the measure of network-wide traffic. Therefore, how to infer network-wide traffic from partial traffic data is extremely important. In this article, a transforms-based Bayesian tensor completion (TBTC) method is proposed to infer network traffic data. First, the heterogeneous network traffic data with missing entries are organized into observation tensors according to temporal dimensions and other attributes. Second, the sparse hierarchical prior is used to induce lateral slices sparsity of factor tensors, which makes the tubal rank of the observation tensor can be estimated. Further, a variational Bayesian inference method is developed for model learning, and an efficient updating method is presented. Finally, two cases of the linear transforms-based tensor completion model are implemented in the experiments. Experimental results on two real-world network traffic datasets validate that the proposed method can efficiently and accurately recover network traffic data.**

***Index Terms*—Network traffic measurement, tensor completion, sparse Bayesian learning, variational Bayesian inference.**

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I. Introduction

As THE bedrock of the next-generation network system, traffic measurement has received increasing attention [1], [2]. It measures the traffic data from the physical layer to the application layer from the network-wide perspective and provides a data foundation for network anomaly monitoring, load balancing, intrusion detection, traffic engineering, *etc*. For this reason, it is particularly important to measure traffic data from a network-wide perspective [3].

In recent years, with the rapid development of the 5 G network, Internet of Things (IoT) and artificial intelligence (AI) technologies, the demand for mobile devices and network services has grown rapidly [4], [5], [6], [7]. At the same time, network traffic data is also showing explosive growth, which brings huge challenges to network-wide traffic measurement. However, the measurement of large network traffic needs a huge overhead [3]. Therefore, how to infer network-wide traffic data from part of the network data has become the focus of the next-generation network systems.

Some studies have shown that end-to-end network measurement data such as traffic, delay and other network information have hidden spatial and temporal correlations [8]. This drives some research on network measurement traffic data inference based on Spatio-temporal correlations [9], [10]. At the same time, these methods can infer other network traffic data by using a few randomly monitoring network traffic data. Therefore, inferring the network-wide traffic data from part of the network traffic data can be considered a missing data recovery task [3].

In addition, some studies have focused on network load balancing [11], confident coverage [12], [13], and communication optimization [14], [15], [16], *etc*, which usually require complete network traffic data. However, the traffic data is often missing and corrupted during the collection due to various factors such as network transmission interruption, network attacks, and signal coverage problems. It seriously affects the monitoring and acquisition of network-wide traffic data. To improve the quality of traffic measurement data and better support the next-generation network systems, recovery of missing traffic data has become one of the great challenges in network science and engineering.

Recently, a class of matrix completion methods was proposed for missing traffic measurement data recovery [10], [17]. However, matrix-based completion methods for traffic data recovery destroy the original structure and correlation. As a multi-dimensional data representation tool, tensor has great advantages to represent multi-dimension and heterogeneous data in network systems [18], [19]. Tensor completion, as one of the important branches of tensor analysis, aims to fill in the missing entries of the observed data and can be regarded as a multi-dimensional data recovery task. Therefore, the recovery of missing network traffic measurement data based on tensor completion has received extensive attention [3], [9]. In [20], an outliner-robust tensor completion method is proposed to recover network traffic data concurrently with network anomaly detection. For highly skewed with heavy tails traffic data, Xie, et al. [21] proposed an expectile tensor completion method to recover missing data. However, these tensor completion methods suffer from low efficiency or low accuracy. To address all challenges above under a unified framework, in this work, a transforms-based Bayesian tensor completion method is proposed to recover missing network traffic measurement data. The main contributions of this article are summarized as follows:

- Heterogeneous traffic measurement data with missing entries are represented by observation tensors according to temporal dimensional and other attributes, which effectively preserves the data structure and correlation. A Bayesian tensor completion method is proposed to recover missing network traffic data. The observation tensor is decomposed into two small factor tensors by linear transforms-based T-SVD. By sparse hierarchical prior and Bayesian inference, the factor tensors obtain lateral slices sparsity, which makes the tubal rank of the observation tensor can be estimated during the tensor completion.
- By diagonalization of the block matrix, an efficient variational Bayesian inference algorithm is presented in the arbitrary invertible linear transforms domain, which effectively reduces the algorithm complexity and improves the missing traffic measurement data recovery efficiency.
- Our proposed Bayesian tensor completion model can be extended to arbitrary invertible linear transforms and is not limited to a specific linear transform. In the experiment, we implement two cases of linear transforms-based (including Cosine-transforms and Random Orthogonal Matrix (ROM) transforms) Bayesian tensor completion method.

This article is structured as follows. Section II presents background and related work. Some notations and basic definitions are introduced in Section III. In Section IV, the details of our proposed model including the probabilistic model, priors, approximate Bayesian inference, and an efficient updating method in arbitrary invertible linear transforms domain are presented. The experiments on the performance of our model under different linear transforms are discussed in Section V. Finally, this article is summarized in Section VI.

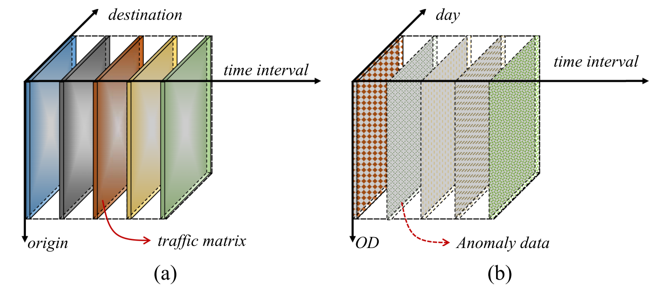

Fig. 1. Illustration of different tensor representations of network traffic data.

II. Background and Related Work

In this section, we will briefly introduce some relevant background information, including tensor and representing traffic data as tensors. In addition, we also introduce some related works of tensor completion for network traffic data recovery.

A. Tensor and Representing Traffic Data as Tensor

Tensor (multidimensional arrays) [22], as an effective tool for multi-dimensional and heterogeneous data representation and calculation, is widely applied in the real world. For example, a hyperspectral image can be regarded as a third-order tensor by its *channels*, *width*, and *height*; a color video can be viewed as a fourth-order tensor indexed by $width \times height \times channels \times frames$; the time-varying urban traffic data can also be represented as a tensor indexed by road-ID, time interval and day; manufacturing process data can be represented as a third-order tensor based on product ID, process parameters and timestamp. Similarly, tensors are also applied in network traffic engineering. For example, as shown in Fig. 1, the time-varying network traffic data can be represented as a third-order tensor indexed by origin, destination, and times interval; the network anomaly data can be viewed as a third-order tensor by its OD (Origin and Destination pair), time interval and days.

B. Related Work

In this subsection, we review the related work on matrix completion, tensor completion, and their application in the recovery of missing network traffic data. In addition, we also briefly identify the differences between our work and existing work.

Traffic data at a certain time can be naturally represented as a matrix according to its source ID and destination ID. Therefore, matrix completion is often used to recover missing traffic data. A matrix completion-based method for estimating missing network latency data for personal devices is proposed in [17]. Based on correlations Spatio-temporal of internet traffic data, a compressive sensing-based missing data recovery method for traffic matrix is proposed in [10]. A fast network monitoring data matrix completion method for acceleration on GPU is proposed in [23]. With the subspace-based matrix completion, a low-cost online network traffic measurement data prediction method is proposed by Jin, et al. [24]. However, for the traffic data with time series, the matrix-based data representation makes the matrix dimension imbalanced, which is not enough to capture the correlations inside the traffic data, and also reduces the performance of the algorithm.

Tensor completion, as one of the important tasks in tensor analysis, has received a lot of attention and has also been applied in many different fields such as internet traffic data recovery [18], network anomaly detection [20], Spatio-temporal data imputation [25], and visual data inpainting [26], *etc*. For traffic data recovery, a novel sequential tensor completion method is proposed to improve computing efficiency and capture the temporal correlation of traffic data [27]. Considering the Spatio-temporal correlation of traffic data, a low-rank tensor factorization-based tensor completion method is proposed to estimate the missing traffic data [9]. A tripartite graph aided tensor completion method is proposed for sparse network measurement data recovery [28]. However, based on CANDECOMP/PARAFAC (CP) [22] decomposition, these methods have the problem that the rank is difficult to estimate. Recently, a class of tensor completion methods based on Bayesian learning with automatic rank estimation has also received some attention. In [29], a hierarchical full Bayesian probabilistic model is proposed for tensor completion based on the CP factorization. Similarly, a Bayesian Gaussian CP tensor decomposition approach for the missing traffic data recovery method is proposed in [25]. However, the model inference method based on Gibbs sampling makes the model converge slowly and is hard to be applied to large datasets.

To the best of our knowledge, we are the first to combine Bayesian learning and tensor completion to recover missing network traffic data, which effectively mines the hidden structural information inside the traffic data and improve accuracy. For the efficiency of the algorithm, a method for posterior estimation in arbitrary invertible linear transformation domain is developed, which reduces the time overhead of the model. Furthermore, with the benefit of sparse Bayesian learning, our model can effectively estimate the rank of traffic data tensors.

III. Notations and Preliminaries

In this section, we summarize the commonly used notations and briefly introduce some basic definitions and tensor operations in this article.

A. Notations

In this article, we denote the fields of real numbers as $\mathbb{R}$. The lowercase, uppercase, and calligraphic letters $(x, X, \mathcal{X})$ are denoted by vectors, matrices, and tensors, respectively. $\mathcal{X}_{ijk}$, $x^{(ij)}$ denotes the (i,j,k)th entry and (i,j)th tube fibers of tensor $\mathcal{X}$. $\mathcal{X}^{\dagger}$, $\hat{\mathcal{X}}$, and $\tilde{\mathcal{X}}$ denote the tensor transport, cosine transform, and arbitrary invertible linear transforms of tensor $\mathcal{X}$ along with the third order. $\odot$ and $\circ$ denote the Face-wise product [30] and Hadamard product. The expectation of latent variables is denoted by $\mathbb{E}(\cdot)$. The operation $\mathrm{diag}(X)$ denotes a vector for the main diagonal of matrix X. The identity matrix and identity tensor are denoted as I, $\mathcal{I}$, respectively. The notations used in this article are summarized in Table I.

TABLE I
Notations

Notation	Description
$\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$	third-order tensor whose size is $n_1 \times n_2 \times n_3$
$\hat{\mathcal{X}}$	the cosine transform of $\mathcal{X}$ along the third-order
$\tilde{\mathcal{X}}$	the arbitrary invertible linear transform of $\mathcal{X}$
$\mathcal{X}^{\dagger} \in \mathbb{R}^{n_2\times n_1\times n_3}$	the transpose tensor $\mathcal{X}$ whose size is $n_2 \times n_1 \times n_3$
$X^{(k)} \in \mathbb{R}^{n_1\times n_2}$	the kth frontal slices of tensor $\mathcal{X}$
$x^{(ij)} \in \mathbb{R}^{n_3}$	the (i,j)th tube fibers of tensor $\mathcal{X}$
$\mathcal{X}_{ijk}$	the (i,j,k)th entry of tensor $\mathcal{X}$
$\overrightarrow{\mathcal{X}_{i::}} \in \mathbb{R}^{1\times n_2\times n_3}$	the ith horizontal slices of tensor $\mathcal{X}$
$\overrightarrow{x_{i::}} = \mathrm{vec}(\mathcal{X}(i,:,:))$	the vectorization ith horizontal slices of tensor $\mathcal{X}$;
$\mathbb{E}(x)$	the expectation of the x
I_{n_3}	the identity matrix with the size of $n_3 \times n_3$
$*_c$	the cosine transform product [30]
$*_L$	the tensor-tensor product with invertible linear transform L
$\mathcal{E}$	the third-order tensor, all of its entry are one
$\mathcal{O}$	the binary tensor, with the size of $n_1 \times n_2 \times n_3$
$\lVert\mathcal{X}\rVert_*$	the tensor nuclear norm of $\mathcal{X}$
$\mathrm{bdiag}(\mathcal{X})$	the block diagonal matrix for third-order tensor $\mathcal{X}$

B. Basic Definitions

This subsection introduces some basic definitions. For a tensor $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$, the *bvec*-operation is forming a $n_1 n_3 \times n_2$ matrix by the frontal slices of tensor $\mathcal{X}$, leading to

$$\mathrm{bvec}(\mathcal{X}) = \left[X^{(1)}; X^{(2)}; \cdots; X^{(n_3)}\right] \in \mathbb{R}^{n_1 n_3\times n_2}.$$

Definition III.1: **(Face-wise product [30])** The face-wise product of tensors $\mathcal{A} \in \mathbb{R}^{n_1\times r\times n_3}$ and $\mathcal{B} \in \mathbb{R}^{r\times n_2\times n_3}$ can be denoted by $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3} = \mathcal{A} \odot \mathcal{B}$, i.e.,

$$X^{(i)} = A^{(i)} B^{(i)}, i = 1, \ldots, n_3. \tag{1}$$

Definition III.2: **(Block diagonal matrix)** Given a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$, we define its block diagonal matrix as

$$\hat{X} = \mathrm{bdiag}(\hat{\mathcal{X}}) = \begin{bmatrix} \hat{X}^{(1)} & & & \\ & \hat{X}^{(2)} & & \\ & & \ddots & \\ & & & \hat{X}^{(n_3)} \end{bmatrix},$$

where $\hat{X} \in \mathbb{R}^{n_1 n_3\times n_2 n_3}$, *bdiag*-operation maps $\hat{\mathcal{X}}$ to $\hat{X}$.

Definition III.3: **(*Mat* -operation)** If a tensor $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$, we will use $\mathrm{mat}(\mathcal{X})$ to denote the following block Toeplitz-plus-Hankel matrix, which is

$$\mathrm{mat}(\mathcal{X}) = \underbrace{\begin{bmatrix} X^{(1)} & X^{(2)} & \cdots & X^{(n_3)} \\ X^{(2)} & X^{(1)} & \cdots & X^{(n_3-1)} \\ \vdots & \vdots & \ddots & \vdots \\ X^{(n_3)} & X^{(n_3-1)} & \cdots & X^{(1)} \end{bmatrix}}_{\text{Block Toeplitz}}$$

$$+ \underbrace{\begin{bmatrix} X^{(2)} & \cdots & X^{(n_3)} & \mathbf{0} \\ \vdots & \iddots & \iddots & \mathcal{X}^{(n_3)} \\ X^{(n_3)} & \mathbf{0} & \iddots & \vdots \\ \mathbf{0} & X^{(n_3)} & \cdots & X^{(2)} \end{bmatrix}}_{\text{Block Hankel}},$$

Algorithm 1: *Ten*-Operation.

Input: A matrix $M \in \mathbb{R}^{n_1 n_3 \times n_2 n_3}$, n_1, n_2, n_3
Output: A tensor $\mathcal{X}$ with the size of $n_1 \times n_2 \times n_3$
$X^{(n_3)} \longleftarrow$ bottom left $n_1 \times n_2$ block of M.
for $k = n_3 - 1, \ldots, 1$ **do**
  $X^{(k)} \longleftarrow$ ([kth block of first (block) column of M] - $X^{(k+1)}$).
end

where the $X^{(k)}$ is the kth frontal faces of $\mathcal{X}$, $\mathbf{0}$ denotes the zero matrices whose size is $n_1 \times n_2$. It is worth noting that the *mat*-operation is invertible, the inverse operation denoted by $\mathrm{ten}(\cdot)$. Algorithm 1 presents the specific implementation of *ten*-operation [30].

C. Cosine Transform Product

This subsection introduces Discrete Cosine Transformation (DCT) and cosine transform product, which plays an important role in our proposed model. The discrete cosine transformation on a vector $v \in \mathbb{R}^n$ can be defined by

$$\bar{v} = C_n v \in \mathbb{R}^n, \tag{2}$$

where C_n is the discrete cosine transform matrix, whose size is $n \times n$ and entries are as follow

$$(C_n)_{ij} = \sqrt{\frac{2 - \delta_{i1}}{n}} \cos\left(\frac{(i-1)(2j-1)\pi}{2n}\right), \quad 1 < i, j < n,$$

where δ_{i1} is the Kronecker delta.

Definition III.4: **(Cosine transform product [30])** Cosine transform product is similar to T-product in [31], which is operation between two third-order tensors. Let $\mathcal{A} \in \mathbb{R}^{m \times r \times h}$ and $\mathcal{B} \in \mathbb{R}^{r \times n \times h}$, cosine transform product can be defined as

$$\mathcal{X} = \mathcal{A} *_c \mathcal{B} = \mathrm{ten}\left(\mathrm{mat}(\mathcal{A})\, \mathrm{mat}\left(\mathcal{B}\right)\right) \in \mathbb{R}^{m \times n \times h}. \tag{3}$$

The cosine transform product in the original domain is present in above. As we know, T-product can be calculated efficiently without explicitly forming the block matrices by fast Fourier transform. Whether exist a similar transformation that can make the cosine transform product more efficient and concise?

For a vector $v \in \mathbb{R}^n$, we define *th*-operation (Toeplitz-plus-Hankel matrix) similar to *mat*-operation as

$$\mathrm{th}(v) = \underbrace{\begin{bmatrix} v_1 & v_2 & \cdots & v_n \\ v_2 & v_1 & \cdots & v_{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ v_n & v_{n-1} & \cdots & v_1 \end{bmatrix}}_{\text{Toeplitz matrix}} + \underbrace{\begin{bmatrix} v_2 & \cdots & v_n & 0 \\ \vdots & \iddots & \iddots & v_n \\ v_n & 0 & \iddots vv & \vdots \\ 0 & v_n & \cdots & v_2 \end{bmatrix}}_{\text{Hankel matrix}}.$$

It is known that $C_n\, \mathrm{th}(v) C_n^{-1} = D$ from [32], where D is a diagonal matrix, which can be computed from the DCT of the first column of $\mathrm{th}(v)$. More specifically, $D = W^{-1}\, \mathrm{diag}(C_n(\mathrm{th}(v)\vec{e}_1))$, where $W = \mathrm{diag}(C_n \vec{e}_1)$, $\vec{e}_1$ is the basis vector. By the definition of the Toeplitz-plus-Hankel matrix, we can find that $\mathrm{th}(v)\vec{e}_1 = (I + Z)v$, where Z is the circulant upshift matrix[1]. Therefore, we can get a more concise calculation method as $D = W^{-1}\, \mathrm{diag}(C_n(I + Z)v)$.

Definition III.5: **(Cosine-transform)** Given a vector $v \in \mathbb{R}^n$, we can define a linear transformation Φ as

$$\Phi(v) = W^{-1} C_n (I + Z) v. \tag{4}$$

In this article, the Φ is defined as the cosine-transform. In addition, this transform is not a direct DCT, but based on DCT. The Φ is an invertible linear transformation, and the inverse operation can be described as

$$\Phi^{-1}(v) = (I + Z)^{-1} (C_n^{-1}(W v)). \tag{5}$$

For simplicity, we denote $\hat{v} = \Phi(v)$. Similarly, for a tensor $\mathcal{X}$, linear transformation Φ can be described as

$$\hat{\mathcal{X}} = \Phi(\mathcal{X}) = \Phi(x^{(ij)}) = W^{-1} C_n (I + Z)\, \mathrm{vec}(x^{(ij)}). \tag{6}$$

According to the above description, we can obtain the concise and efficient form of cosine transform product by a linear transformation Φ, which is

$$\mathcal{X} = \mathcal{A} *_c \mathcal{B} = \Phi^{-1}(\Phi(\mathcal{A}) \odot \Phi(\mathcal{B})). \tag{7}$$

For cosine transform product, from the transform domain, we can obtain more brief representation as follow

$$\hat{X}^{(k)} = \hat{A}^{(k)} \hat{B}^{(k)}; k = 1, \ldots, n_3. \ \ , \tag{8}$$

where $\hat{X}$ denotes block diagonal matrix [33]. It is known that $C_n\, \mathrm{th}(v) C_n^{-1} = \mathrm{diag}(\Phi(v))$, then, we have

$$(C_{n3} \otimes I_{n_1})\, \mathrm{mat}(\mathcal{X}) \left(C_{n_3}^{-1} \otimes I_{n_2}\right) = \hat{X}. \tag{9}$$

IV. Transforms-Based Bayesian Tensor Completion Algorithm

In this section, a transforms-based Bayesian tensor completion algorithm is proposed as follows. First, the probabilistic model and priors based on the cosine transform product are presented. Second, a method is developed for estimating factor tensors based on Bayesian inference. Third, through the tensor-tensor product with arbitrary invertible linear transforms theory, our proposed model can be naturally generalized to arbitrary invertible linear transforms. Finally, an efficient updating method in arbitrary invertible linear transforms domain is developed.

In addition, Fig. 2 shows the main process in our proposed model for tensor completion, including tensor arbitrary invertible linear transforms for incomplete tensor, tensor low-tubal-rank model for tensor factorization, variational Bayesian inference for model learning to induce the lateral slices sparsity of factor tensors, and tensor-tensor product with arbitrary invertible linear transforms.

A. Probabilistic Model and Priors

In this article, the tensor completion problem is considered in a probabilistic framework, all unknown observed data and quantities are treated as random variables and their joint distributions are specified. Let $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ be an incomplete third-order

[1] In Matlab, $Z = \mathrm{diag}(\mathrm{ones}(n-1,1),1)$.

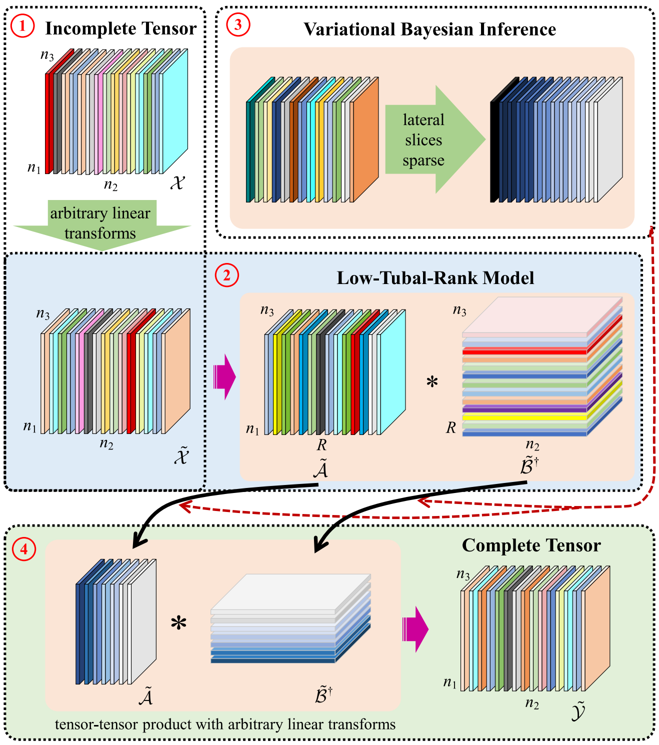

Fig. 2. Graphical illustration of the main process of our proposed model for tensor completion. (1) arbitrary invertible linear transforms for incomplete tensor $\mathcal{X}$; (2) low-tubal-rank model for tensor factorization; (3) lateral slices sparsity of factor tensors induced by variational Bayesian inference; (4) tensor-tensor product with arbitrary invertible linear transforms.

tensor. $\mathcal{X}_{ijk}$ is observed entry if $(i,j,k) \in \Omega$, where Ω denotes a set of indices. The $\mathcal{O}$ is a binary tensor that indicates which entries of $\mathcal{X}$ are observed.

1) Optimization Model: Based on the low-tubal-rank model [33], our proposed tensor completion model can be described as

$$\mathcal{X} = \mathcal{A} *_c \mathcal{B}^{\dagger}, \tag{10}$$

where factor tensors $\mathcal{A} \in \mathbb{R}^{n_1 \times R \times n_3}$, $\mathcal{B} \in \mathbb{R}^{n_2 \times R \times n_3}$ and $*_c$ denotes the cosine transform product. The incomplete low-rank tensor recovering, which aims to exactly recover a low-rank tensor from incomplete observation, can be taken as solving the affine rank minimization problem. Its mathematical model can be written as

$$\begin{aligned} &\text{minimize} \quad \operatorname{rank}(\mathcal{X}) \\ &\text{subject to} \quad \mathcal{Y}_\Omega = \mathcal{X} \circ \mathcal{O}, \end{aligned} \tag{11}$$

where the $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an unknown low-rank tensor, $\circ$ denotes the tensor Hadamard product and $\mathcal{Y}_\Omega$ is the observed tensor. However, minimizing the rank of a tensor is an NP-hard problem [34], and no-know polynomial-time algorithms exist. To overcome this difficulty, an approximate Tensor Nuclear Norm (TNN) [35] based method is commonly used to solve, which is given by

$$\begin{aligned} &\text{minimize} \quad \|\mathcal{X}\|_* \\ &\text{subject to} \quad \mathcal{Y}_\Omega = \mathcal{X} \circ \mathcal{O}. \end{aligned} \tag{12}$$

To optimize the problem in (12), the Bayesian theory is used as follows. Since a low-tubal-rank estimate of tensor $\mathcal{X}$ is sought, we should achieve lateral slices sparsity of factor tensors $\mathcal{A}$ and $\mathcal{B}$. In order to fit sparsity-inducing prior distribution, we associate each tube of factor tensors $\mathcal{A}$ and $\mathcal{B}$ with Gaussian priors of λ, which can be described as follows

$$p(\mathcal{A}|\lambda) = \prod_{i=1}^{n_1} \prod_{r=1}^{R} \mathcal{N}\left(a^{(ir)} | \mathbf{0}, \lambda_r^{-1} I_{n_3}\right), \tag{13}$$

$$p(\mathcal{B}|\lambda) = \prod_{j=1}^{n_2} \prod_{r=1}^{R} \mathcal{N}\left(b^{(jr)} | \mathbf{0}, \lambda_r^{-1} I_{n_3}\right), \tag{14}$$

where $a^{(ir)}$ and $b^{(jr)}$ denote the (i,r)th and (j,r)th tube of factor tensors $\mathcal{A}$ and $\mathcal{B}$, I_{n_3} denotes the identity matrix. In addition, to fit the probabilistic model, we use the Gamma hyperprior to describe λ, as follow

$$p(\lambda) = \prod_{r=1}^{R} \operatorname{Gamma}\left(\lambda_r | a_0^{\lambda}, b_0^{\lambda}\right), \tag{15}$$

where $\forall r \in [1, R]$, the λ_r follows an independent Gamma distribution, $\operatorname{Gamma}(x|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x > 0$ denotes the Gamma distribution, $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$ is the Gamma function. Then, the expectation of the latent variable x can be described as $\mathbb{E}[x] = \dfrac{\alpha}{\beta}$.

2) Noise and Observation Models: Noise is inevitable during data acquisition, to better meet the demand of recovering missing data in practical applications and improve the robustness of the model, the observation model is generated according to

$$\mathcal{Y}_\Omega = P_\Omega(\mathcal{X} + \mathcal{N}), \tag{16}$$

where $\mathcal{Y}_\Omega$ is the observation tensor with noise, $\mathcal{N}$ is the noise term.

According to the above description, the white Gaussian noise is incorporated into the observation model, then, we can obtain conditional distribution for each tube of the observed tensor $\mathcal{Y}_\Omega$, which is

$$p(\mathcal{Y}_\Omega | \mathcal{A}, \mathcal{B}, \tau) = \prod_{ij} \mathcal{N}\left(y^{(ij)} | \overrightarrow{\mathcal{A}_{i::}} *_c \overrightarrow{\mathcal{B}_{j::}}^{\dagger}, \tau^{-1} I_{n_3}\right). \tag{17}$$

To complete the fully Bayesian treatment in our proposed model, the noise entries are assumed to be i.i.d random variables obeying Gaussian distribution, where precision is τ and mean is zero. Further, we place a conjugate prior over the noise precision as

$$p(\tau) = \operatorname{Gamma}\left(\tau | c^{\tau}, d^{\tau}\right). \tag{18}$$

Finally, the probabilistic graphical model of Bayesian tensor completion is shown in Fig. 3. According to the model described above, the joint distribution of $p(\mathcal{Y}_\Omega, \mathcal{S})$ can be written as follow

$$p(\mathcal{Y}_\Omega, \mathcal{S}) = p(\mathcal{Y}_\Omega | \mathcal{A}, \mathcal{B}, \tau) p(\mathcal{A}|\lambda) p(\mathcal{B}|\lambda) p(\lambda) p(\tau), \tag{19}$$

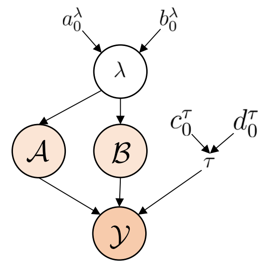

Fig. 3. Graphical illustration of our proposed tensor completion model.

where $\mathcal{S} = \{\mathcal{A}, \mathcal{B}, \lambda, \tau\}$ is the collection of all of the unknown hyper-parameters and latent variables in our proposed Bayesian model.

Through the analysis above, the aim of our proposed model is to estimate the full posterior distribution of $\mathcal{S}$ from the observed data $\mathcal{Y}_\Omega$, which can be described as follow

$$p\left(\mathcal{S}|\mathcal{Y}_\Omega\right) = \frac{p\left(\mathcal{S}, \mathcal{Y}_\Omega\right)}{\int p\left(\mathcal{S}, \mathcal{Y}_\Omega\right) d\mathcal{S}}. \tag{20}$$

Therefore, missing entries can be predicted based on the posterior distribution of $\mathcal{S}$, which is

$$p\left(\mathcal{Y}_{\backslash\Omega}|\mathcal{Y}_\Omega\right) = \int p\left(\mathcal{Y}_{\backslash\Omega}|\mathcal{S}\right) p\left(\mathcal{S}|\mathcal{Y}_\Omega\right) d\mathcal{S}, \tag{21}$$

where $\mathcal{Y}_{\backslash\Omega}$ denotes the predictive missing entries from incomplete observed tensor $\mathcal{Y}_\Omega$.

B. Model Learning Via Bayesian Inference

Actually, the marginalizing of all variables of $p(\mathcal{Y}_{\backslash\Omega})$ can't be computed directly. Therefore, the inference joint distribution in (19) by Bayesian theory is intractable. For this problem, the maximum a posteriori (MAP), variational Bayesian inference, and other approximate estimation methods are used to solve frequently.

In this subsection, a deterministic approximate inference method for the probabilistic tensor completion model learning is developed under the variational Bayesian inference framework. Therefore, our objective is to find a distribution $q(\mathcal{S})$ to approximate the posterior distribution $p(\mathcal{S}|\mathcal{Y}_\Omega)$ by minimizing the Kullback-Leibler (KL) divergence, which can be described as

$$\begin{aligned} \mathrm{KL}\left(q(\mathcal{S}) \| p\left(\mathcal{S}|\mathcal{Y}_\Omega\right)\right) &= \int q(\mathcal{S}) \ln \frac{q(\mathcal{S})}{p\left(\mathcal{S}|\mathcal{Y}_\Omega\right)} d\mathcal{S} \\ &= \ln p\left(\mathcal{Y}_\Omega\right) - \mathcal{L}(q), \end{aligned} \tag{22}$$

where $\ln p(\mathcal{Y}_\Omega)$ represents our model evidence, $q(\mathcal{S})$ is probability density function, and $\mathcal{L}(q)$ represents the lower bound of $\ln p(\mathcal{Y}_\Omega)$, which is defined by

$$\mathcal{L}(q) = \int q(\mathcal{S}) \ln \frac{p\left(\mathcal{Y}_\Omega, \mathcal{S}\right)}{q(\mathcal{S})} d\mathcal{S}. \tag{23}$$

Since the model evidence $\ln p(\mathcal{Y}_\Omega)$ is a constant, minimizing the KL divergence is equal to maximizing the lower bound of $\mathcal{L}(q)$, the maximum of the lower bound occurs when the KL divergence is equal to zero, which implies that $\ln p(\mathcal{Y}_\Omega)$ is equal to $\mathcal{L}(q)$.

For tractable inference under variational Bayesian framework and based on mean-field approximation, it is assumed that all of the latent variables in $q(\mathcal{S})$ can be factorized as

$$q(\mathcal{S}) = q(\mathcal{A})q(\mathcal{B})q(\lambda)q(\tau). \tag{24}$$

It means that all of the latent variables in $\mathcal{S}$ are independent, therefore, we can update each latent variable by keeping others fixed. The posterior of each latent variable approximation $q(\mathcal{S}_i)$ can be obtained as follow

$$\ln q\left(\mathcal{S}_i\right) = \mathbb{E}_{\mathcal{S}\backslash\mathcal{S}_i}\left[\ln p(\mathcal{Y}_\Omega, \mathcal{S})\right] + \text{ const }, \tag{25}$$

where the $\mathcal{S}_i$ denotes the ith latent variable in $\mathcal{S}$. The expectation $\mathbb{E}_{\mathcal{S}\backslash\mathcal{S}_i}[\cdot]$ denotes the variational distribution of all latent variables in set $\mathcal{S}$ expect $\mathcal{S}_i$. Since all distributions in our model given above belong to the conjugate exponential family, it is easy to find a posterior approximation for each latent variable.

1) Inference for Factor Tensors $\mathcal{A}$ and $\mathcal{B}$: From the probabilistic graph model shown in Fig. 3 and observation modeling join distribution in (19), we can find the inference of the factor tensor $\mathcal{A}$ can be performed by receiving the messages from observed data and its co-parents, including hyper-parameter τ, and factor tensor $\mathcal{B}$. By applying the (25), and combining the prior in (13) and observation model in (17), the posterior of $q(\mathcal{A})$ can be found as a Gaussian distribution as

$$q(\mathcal{A}) = \prod_{i=1}^{n_1} \mathcal{N}\left(\overrightarrow{a_{i::}} \mid \mathbb{E}\left[\overrightarrow{a_{i::}}\right], \Sigma^{\mathrm{A}}\right), \tag{26}$$

whose posterior parameters are given by

$$\mathbb{E}\left[\overrightarrow{a_{i::}}\right] = \mathbb{E}\left[\tau\right] \Sigma^{\mathrm{A}} \mathrm{mat}\left(\mathbb{E}\left[\mathcal{B}\right]\right)^\dagger \overrightarrow{y_{i::}}, \tag{27}$$

$$\Sigma^{\mathrm{A}} = \left(\mathbb{E}\left[\tau\right] \mathbb{E}\left[\mathrm{mat}(\mathcal{B})^\dagger \mathrm{mat}(\mathcal{B})\right] + \mathbb{E}\left[\Gamma\right]\right)^{-1}. \tag{28}$$

Similar to the inference of factor tensor $\mathcal{A}$, by combining the prior in (14) and observation model in (17), the posterior $q(\mathcal{B})$ can be obtained as

$$q(\mathcal{B}) = \prod_{j=1}^{n_2} \mathcal{N}\left(\overrightarrow{b_{j::}} \mid \mathbb{E}\left[\overrightarrow{b_{j::}}\right], \Sigma^{\mathrm{B}}\right), \tag{29}$$

with mean and covariance as

$$\mathbb{E}\left[\overrightarrow{b_{j::}}\right] = \mathbb{E}\left[\tau\right] \Sigma^{\mathrm{B}} \mathrm{mat}\left(\mathbb{E}\left[\mathcal{A}\right]\right)^\dagger \overrightarrow{y_{:j:}}, \tag{30}$$

$$\Sigma^{\mathrm{B}} = \left(\mathbb{E}\left[\tau\right] \mathbb{E}\left[\mathrm{mat}(\mathcal{A})^\dagger \mathrm{mat}(\mathcal{A})\right] + \mathbb{E}\left[\Gamma\right]\right)^{-1}, \tag{31}$$

where $\mathbb{E}[\Gamma] = \mathrm{mat}(\mathrm{diag}(\mathbb{E}[\lambda_1], \mathbb{E}[\lambda_2], \ldots, \mathbb{E}[\lambda_R]))$.

2) Inference for Hyper-Parameters λ: Similarly to the above, the inference of λ can be performed by incorporating messages from its hyperprior and receiving messages from factor tensors $\mathcal{A}$ and $\mathcal{B}$. By applying (25), the posterior of λ_r follows an independent Gamma distribution, which can be described as

$$q(\lambda) = \prod_{r=1}^{R} \mathrm{Gamma}\left(\lambda_r \mid a_r^\lambda, b_r^\lambda\right), \tag{32}$$

the expectation of λ can be obtained by $\mathbb{E}[\lambda] = \frac{a_r^\lambda}{b_r^\lambda}$.

Algorithm 2: Bayesian Tensor Completion Model Under $*_c$.

Input: The observed tensor $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and index set Ω

Initialize $\hat{\mathcal{A}}, \hat{\mathcal{B}}$ by low-tubal-rank model via Eq. (10).
Initialize: $\lambda, \tau, \Sigma^{\mathrm{A}}, \Sigma^{\mathrm{B}}$;
repeat
  Update the covariance Σ^{A} via Eq. (28);
  Update the mean $\mathbb{E}\left[\overrightarrow{a_{i::}}\right]$ via Eq. (27);
  Update the $q(\mathcal{A})$ via Eq. (26);
  Update the covariance Σ^{B} via Eq. (31);
  Updata mean $\mathbb{E}\left[\overrightarrow{b_{j::}}\right]$ via Eq. (30);
  Update the $q(\mathcal{B})$ via Eq. (29);
  Update the λ via Eq. (32);
  Update the noise precision τ via Eq. (33);
  Reduce the tubal rank by removing zero-components of lateral slices of factor tensor $\mathcal{A}$ and $\mathcal{B}$;
until *minimum tensor nuclear norm*;

3) Inference for Noise Precision τ: Similarly to inference λ, the noise precision τ can be performed by incorporating the information from its hyperprior and receiving messages from the observed data. By applying (25), the posterior of noise τ obeys a Gamma distribution, which can be described as

$$q(\tau) = \mathrm{Gamma}(\tau | c^{\tau}, d^{\tau}), \tag{33}$$

the expectation of τ can be obtained by $\mathbb{E}(\tau) = \dfrac{c^{\tau}}{d^{\tau}}$.

The Bayesian tensor completion model under $*_c$ can be summarized as follows: First, estimate the factor tensors $\mathcal{A}$ and $\mathcal{B}$ by (26) and (29), whose covariance is updated by (28) and (31), and mean is updated via (27) and (30). Second, the posterior distribution of hyper-parameters λ is updated via (32). Finally, the posterior distribution of noise precision τ is updated via (33), until convergence. For clearly, the Bayesian tensor completion model under $*_c$ is summarized in algorithm 2.

C. Arbitrary Invertible Linear Transforms-Based Bayesian Tensor Completion Model

The cosine transform product is introduced in Definition III.4, as well as the t-product, which can replace the complex block-structured matrix operation in the origin domain with a specific invertible linear transform. In practical applications, the algorithm implementation of t-product and cosine transform product can be performed by matrix product in the transform domain through linear transforms for each tube fibers. Is it possible to define a tensor-tensor product that is not limited to a specific linear transform? If so, can our proposed Bayesian tensor completion model also not be limited to the tensor-tensor product under a specific linear transform?

In this subsection, for these questions, through the definition of tensor-tensor product with an arbitrary invertible linear transform presented in [30], our proposed Bayesian tensor completion model is also extended to arbitrary invertible linear transforms. To better illustrate the extension of our proposed Bayesian tensor completion model under arbitrary invertible linear transforms, the following introduces some basic definitions of tensor computation in arbitrary invertible linear transforms.

Definition IV.1: **(Tensor invertible linear transform)** Consider an arbitrary invertible linear transform $\boldsymbol{L} : \mathbb{R}^{n_1 \times n_2 \times n_3} \rightarrow \mathbb{R}^{n_1 \times n_2 \times n_3}$, then tensor invertible linear transform can be defined as

$$\tilde{\mathcal{X}} = \boldsymbol{L}(\mathcal{X}) = \mathcal{X} \times_3 L, \tag{34}$$

where $\times_3$ is the model-3 product [22], and $L \in \mathbb{R}^{n_3 \times n_3}$ is an arbitrary invertible matrix. Obviously, we can easily get the inverse transform of $\boldsymbol{L}$, which is

$$\boldsymbol{L}^{-1}(\mathcal{X}) = \mathcal{X} \times_3 L^{-1}, \tag{35}$$

where L^{-1} is inverse of matrix L.

Suppose $L = F_{n_3}$, where F_{n_3} is a discrete Fourier transform matrix whose size is $n_3 \times n_3$. Given tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times r \times n_3}$ and $\mathcal{B} \in \mathbb{R}^{r \times n_2 \times n_3}$, t-product [31] between them can be denoted as $\mathcal{X} = \mathcal{A} *_F \mathcal{B}$, which can also be written as $\boldsymbol{L}(\mathcal{X}) = \boldsymbol{L}(\mathcal{A}) \odot \boldsymbol{L}(\mathcal{B})$ in frequency domain. If L is an arbitrary invertible linear matrix, tensor-tensor products with arbitrary invertible linear transforms [30] can be denoted as $\mathcal{X} = \mathcal{A} *_L \mathcal{B}$.

Definition IV.2: **(Identity tensor)** Suppose $\boldsymbol{L}$ is an arbitrary invertible linear transform in (34), the tensor $\mathcal{I} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an identity tensor, if $\mathcal{I}$ satisfies all frontal slices of $\boldsymbol{L}(\mathcal{I})$ are identity matrices.

Definition IV.3: **(Orthogonal tensor [31])** Suppose a tensor $\mathcal{Q} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\boldsymbol{L}$ is an arbitrary invertible linear transform in (34). If $\mathcal{Q}$ satisfies $\mathcal{Q}^{\dagger} * \mathcal{Q} = \mathcal{Q} *_L \mathcal{Q}^{\dagger} = \mathcal{I}$, $\mathcal{Q}$ is defined as an orthogonal tensor.

Definition 4.4: **(Tensor transport)** $\boldsymbol{L}$ is an arbitrary invertible linear transform in (34), the tensor transpose of $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ under $\boldsymbol{L}$ can be denoted as $\mathcal{X}^{\dagger} \in \mathbb{R}^{n_2 \times n_1 \times n_3}$, which satisfies $\boldsymbol{L}(\mathcal{X}^{\dagger})^{(k)} = \boldsymbol{L}(\mathcal{X}^{(k)})^{\dagger}$, $k = 1, \ldots, n_3$.

Definition IV.5: **(Tensor Singular Value Decomposition (T-SVD) [31])** Suppose $\boldsymbol{L}$ is an arbitrary invertible linear transform in (34), tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ can be factorized as

$$\mathcal{X} = \mathcal{U} *_L \Sigma *_L \mathcal{V}^{\dagger}, \tag{36}$$

where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors, $\Sigma \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is the f-diagonal tensor.

Based on the above definition, our model can be naturally generalized to arbitrary invertible linear transforms. The low-tubal-rank model with arbitrary invertible linear transforms can be described as

$$\mathcal{X} = \mathcal{A} *_L \mathcal{B}^{\dagger}, \tag{37}$$

where $\mathcal{A} \in \mathbb{R}^{n_1 \times R \times n_3}$, $\mathcal{B} \in \mathbb{R}^{n_2 \times R \times n_3}$, and $*_L$ denotes the tensor-tensor product with arbitrary invertible linear transforms. At model initialization, the factor tensors can be set as $\mathcal{A}_0 = \mathcal{U}_0 *_L \Sigma_0^{1/2}$, and $\mathcal{B}_0 = \mathcal{V}_0 *_L \Sigma_0^{1/2}$, in arbitrary invertible linear transforms domain $\tilde{A}_0 = \tilde{U}_0 * \tilde{\Sigma}_0^{1/2}$, and $\tilde{B}_0 = \tilde{V}_0 * \tilde{\Sigma}_0^{1/2}$.

D. Efficient Updating Method in Arbitrary Invertible Linear Transforms Domain

Although the algorithm of the Bayesian tensor completion model under $*_c$ has been given in Subsection IV-B, it is still too time-consuming. In this subsection, an efficient updating method for Bayesian tensor completion in arbitrary invertible linear transforms domain is developed.

From (27), we group all the horizontal slices of factor tensor $\mathcal{A}$ together in cosine transform domain can be obtained by

$$\begin{aligned}\text{bvec}(\mathbb{E}[\tilde{\mathcal{A}}]^{\dagger}) &= M \cdot \text{bvec}(\mathbb{E}[\mathcal{A}]^{\dagger})\\ &= M \cdot \mathbb{E}[\overrightarrow{a_{i::}^{\dagger}}, \ldots, \overrightarrow{a_{m::}^{\dagger}}]\\ &= \mathbb{E}[\tau] \cdot M \cdot \tilde{\Sigma}^{\text{A}} \cdot \text{opt}(\mathbb{E}[\mathcal{B}])^{\dagger} \cdot \text{bvec}(\mathcal{Y}^{\dagger}),\end{aligned} \tag{38}$$

where the $M = C_{n_3} \otimes I_R$, M^{-1} is inverse of M, opt$(\cdot)$ means mat$(\cdot)$ operation, with the (9), the block Toeplitz-plus-Hankel matrix mat$(\mathbb{E}[\mathcal{B}])^{\dagger}$ and covariance Σ^{A} can be block diagonalized.

Similarly, for an arbitrary invertible linear transform $\boldsymbol{L}$, $M = L \otimes I_R$. Therefore, we can infer the distribution of factor tensor $\mathcal{A}$ in arbitrary invertible linear transforms domain efficiently. The mean and covariance of kth frontal slice of factor tensor $\mathcal{A}$ in arbitrary linear transforms domain is updated by follows

$$\mathbb{E}\left[\tilde{A}^{(k)}\right] = \mathbb{E}\left[\tau\right] \langle \tilde{X}^{(k)} \rangle_{\mathcal{O}} \mathbb{E}\left[\tilde{B}^{(k)}\right] \tilde{\Sigma}^{A^{(k)}}, \tag{39}$$

$$\tilde{\Sigma}^{A^{(k)}} = \left(\mathbb{E}\left[\tau\right] \mathbb{E}\left[\tilde{B}^{(k)\dagger}\tilde{B}^{(k)}\right] + \mathbb{E}\left[\Gamma_r^{(k)}\right]\right)^{-1}, \tag{40}$$

where $\langle \mathcal{X} \rangle_{\mathcal{O}} = \mathcal{X} \circ \mathcal{O} + \mathcal{Y} \circ (\mathcal{E} - \mathcal{O})$, and the $\mathcal{Y}$ denotes recovery tensor.

Similarly, for inferring the distribution of factor tensor $\mathcal{B}$, the kth frontal slice expectation of $\mathbb{E}[\mathcal{B}]$ and covariance matrix in arbitrary invertible linear transforms domain can be updated by follows

$$\mathbb{E}\left[\tilde{B}^{(k)}\right] = \mathbb{E}\left[\tau\right] \langle \tilde{X}^{(k)} \rangle_{\mathcal{O}} \mathbb{E}\left[\tilde{A}^{(k)}\right] \tilde{\Sigma}^{B^{(k)}}, \tag{41}$$

$$\tilde{\Sigma}^{B^{(k)}} = \left(\mathbb{E}\left[\tau\right] \mathbb{E}\left[\tilde{A}^{(k)\dagger}\tilde{A}^{(k)}\right] + \mathbb{E}\left[\Gamma_r^{(k)}\right]\right)^{-1}, \tag{42}$$

where $\mathbb{E}[\Gamma_r^{(k)}] = \text{diag}(\mathbb{E}[\lambda_1^k], \mathbb{E}[\lambda_2^k], \ldots, \mathbb{E}[\lambda_R^k])$, $\mathbb{E}[\tilde{A}^{(k)}] \in \mathbb{R}^{n_1 \times R}$ and $\mathbb{E}[\tilde{B}^{(k)}] \in \mathbb{R}^{n_2 \times R}$, which denote the kth frontal slice of $\mathbb{E}[\tilde{\mathcal{A}}]$ and $\mathbb{E}[\tilde{\mathcal{B}}]$ respectively. The expectations of $\mathbb{E}[\tilde{B}^{(k)\dagger}\tilde{B}^{(k)}]$ and $\mathbb{E}[\tilde{A}^{(k)\dagger}\tilde{A}^{(k)}]$ can be written as follows

$$\mathbb{E}\left[\tilde{B}^{(k)\dagger}\tilde{B}^{(k)}\right] = \mathbb{E}\left[\tilde{B}^{(k)\dagger}\right] \mathbb{E}\left[\tilde{B}^{(k)}\right] + n_2 n_3 \tilde{\Sigma}^{B^{(k)}}, \tag{43}$$

$$\mathbb{E}\left[\tilde{A}^{(k)\dagger}\tilde{A}^{(k)}\right] = \mathbb{E}\left[\tilde{A}^{(k)\dagger}\right] \mathbb{E}\left[\tilde{A}^{(k)}\right] + n_1 n_3 \tilde{\Sigma}^{A^{(k)}}. \tag{44}$$

Based on the above analysis, for Bayesian tensor completion under $*_c$, we replace (27) and (30) with (39) and (41), which can avoid directly calculating covariance matrices in the original domain, and turn to fast updating in cosine transforms domain. With this, the (39)–(44) are not only limited to the Bayesian tensor completion model under cosine transform, but also are applicable to the Bayesian tensor completion model with arbitrary invertible linear transforms.

Algorithm 3: Transforms-Based Bayesian Tensor Completion (TBTC).

Input: The observed tensor $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and index set Ω

Initialize $\tilde{\mathcal{A}}, \tilde{\mathcal{B}}$ by low-tubal-rank model in arbitrary invertible linear transforms domain via Eq. (37);
Initialize: $\lambda, \tau, \tilde{\Sigma}^{\text{A}}, \tilde{\Sigma}^{\text{B}}$.
repeat
  for $k = 1, \ldots, n_3$ **do**
    Update the covariance of kth frontal slice of $q(\tilde{A})$ via Eq. (40);
    Update the mean of kth frontal slice of $q(\tilde{A})$ via Eq. (39);
    Update the covariance of kth frontal slice of $q(\tilde{B})$ via Eq. (42);
    Update the mean of kth frontal slice of $q(\tilde{B})$ via Eq. (41);
    Update the posterior distribution of hyper-parameters λ via Eq. (45);
  end
  Update the noise precision τ via Eq. (46);
  Reduce the effective tubal rank by removing zero-components of $\tilde{A}^{(k)}, \tilde{B}^{(k)}$;
until *minimum tensor nuclear norm*;

With the (32), we can obtain the posterior parameters in arbitrary invertible linear transforms domain as

$$\begin{aligned}a_r^{\lambda} &= \frac{n_1 + n_2}{2} + a_0^{\lambda},\\ b_r^{\lambda} &= \frac{1}{2n_3} \text{diag}\left(\mathbb{E}\left[\tilde{B}^{(i)\dagger}\tilde{B}^{(i)}\right] + \mathbb{E}\left[\tilde{A}^{(i)\dagger}\tilde{A}^{(i)}\right]\right) + b_0^{\lambda}.\end{aligned} \tag{45}$$

With (33), the posterior parameters c^{τ} and d^{τ} can be updated by

$$\begin{aligned}c^{\tau} &= \frac{1}{2}\sum_{ijk} \mathcal{O}_{ijk} + c_0^{\tau},\\ d^{\tau} &= \frac{1}{2}\sum_{ij} \mathbb{E}\left[\left\| x^{(ij)} - \overrightarrow{\mathcal{A}}_{i::} * \overrightarrow{\mathcal{B}}_{j::}^{\dagger} \right\|^2\right] + d_0^{\tau},\end{aligned} \tag{46}$$

where $\sum_{ij} \mathbb{E}[\| x^{(ij)} - \overrightarrow{\mathcal{A}}_{i::} * \overrightarrow{\mathcal{B}}_{j::}^{\dagger} \|^2]$ is the error measurement of our model.

In summary, based on the algorithm updated by (39)–(46), and the low-tubal-rank model proposed in (37), the Bayesian tensor completion model can be generalized to arbitrary invertible linear transforms domain, not only limited in cosine transforms domain. Algorithm 3 shows the details of TBTC model.

E. Rank Estimation and Parameters Initialization

Based on (32) and (45), the expectation of λ_r is given by $\mathbb{E}(\lambda_r) = a_r^{\lambda}/b_r^{\lambda}$, which depends on $\mathbb{E}[\tilde{B}^{(i)\dagger}\tilde{B}^{(i)}]$ and $\mathbb{E}[\tilde{A}^{(i)\dagger}\tilde{A}^{(i)}]$. While the algorithm iterating, smaller $\text{diag}(\mathbb{E}[\tilde{B}^{(i)\dagger}\tilde{B}^{(i)}] + \mathbb{E}[\tilde{A}^{(i)\dagger}\tilde{A}^{(i)}])$ will lead to a larger $\mathbb{E}(\lambda_r)$, which induces lateral slices sparsity of factor tensors $\mathcal{A}$ and $\mathcal{B}$. With the lateral slices sparsity of factor tensors, it naturally leads to low-tubal-rank of completion tensor. However, in practical application, the main diagonal of Σ is hardly equal to zero, but

TABLE II
RECOVERY RESULTS OF TBTC-COSINE AND TBTC-ROM ON THE SYNTHETIC DATA

Tensor size	Rank	Noise	Missing rate	TBTC-cosine			TBTC-ROM		
$n_1 \times n_2 \times n_3$	R	σ_ε^2	ρ	SRR	MAE	RRSE	SRR	MAE	RRSE
$100 \times 100 \times 10$	5	0	20%	9.36E-01	8.14E-03	4.83E-02	9.58E-01	6.37E-03	3.75E-02
			40%	8.51E-01	1.82E-02	7.68E-02	9.29E-01	1.18E-02	4.93E-02
			60%	7.66E-01	2.84E-02	9.75E-02	8.38E-01	2.17E-02	7.39E-02
			80%	7.31E-01	3.37E-02	1.01E-01	7.54E-01	3.13E-02	9.23E-02
		1E-02	20%	9.36E-01	1.21E-02	4.90E-02	9.57E-01	1.04E-02	3.88E-02
			40%	8.51E-01	2.13E-02	7.70E-02	9.27E-01	1.49E-02	5.02E-02
			60%	7.66E-01	3.05E-02	9.71E-02	8.36E-01	2.39E-02	7.40E-02
			80%	7.29E-01	3.48E-02	1.00E-01	7.52E-01	3.25E-02	9.20E-02
$200 \times 200 \times 20$	10	0	20%	9.44E-01	7.46E-03	4.26E-02	9.76E-01	5.11E-03	2.92E-02
			40%	8.66E-01	1.67E-02	6.79E-02	9.30E-01	1.17E-02	4.75E-02
			60%	7.90E-01	2.58E-02	8.57E-02	8.83E-01	1.85E-02	6.10E-02
			80%	8.16E-01	2.66E-02	7.63E-02	8.34E-01	2.53E-02	7.24E-02
		1E-02	20%	9.43E-01	1.15E-02	4.38E-02	9.75E-01	9.20E-03	3.12E-02
			40%	8.65E-01	1.98E-02	6.81E-02	9.28E-01	1.49E-02	4.83E-02
			60%	7.88E-01	2.80E-02	8.57E-02	8.80E-01	2.06E-02	6.14E-02
			80%	8.13E-01	2.78E-02	7.63E-02	8.31E-01	2.65E-02	7.25E-02

very close to zero. This yield tubal rank is not suitable for our Bayesian tensor completion model. To address this problem, truncated T-SVD is used in our model. By applying truncated T-SVD, some values in Σ those are close to zero can be set to zero, which ensures that the low-tubal-rank model can perform better.

In order to achieve better performance for our proposed model while avoiding getting stuck in a local optimum, it is essential to choose the appropriate initialization parameters. In our model, the top-level hyper-parameters $a_0^\lambda, b_0^\lambda, c_0^\tau, d_0^\tau$ are set to 10^{-5} with model precision $\mathbb{E}[\tau] = c_0^\tau / d_0^\tau = 1$ and $\mathbb{E}[\lambda] = \mathrm{diag}(I)$.

V. EXPERIMENTS AND DISCUSSION

In this section, to evaluate our proposed model for the missing network-wide traffic measurement data completion, we conduct experiments on synthetic data and two real-world network-wide traffic measurement datasets and compare them with other state-of-the-art matrix/tensor completion models, including FBCP [29], BGCP [25], TCTF [26], TMac [36], FGSR [37], and FNN [37]. In particular, the TMac has two versions, TMac-Dec and TMac-Inc, the former uses the decreased rank to obtain the optimal rank, while the latter uses the increased rank. FGSR and FNN are matrix-based methods, which cannot deal with tensor-based data. Therefore, in the experimental part, for FGSR and FNN, we flatten the tensors into matrices according to the third dimension.

Moreover, in our proposed model, we consider two cases of linear transforms: (1) Cosine-transform (Cosine), which is based on the linear transform proposed in Definition III.5; (2) Random Orthogonal Matrix transform (ROM), the transform matrix L is a random orthogonal matrix gotten by QR decomposition. In our experiments, the Successful Recovery Rate (SRR), Mean the Absolute Error (MAE) [27], Root Relative Squared Error (RRSE) [38], and computation time are used as performance metrics, which are defined as follows

$$\mathrm{SRR} = \frac{\sum \eta_{ijk}}{n_1 \times n_2 \times n_3} \text{ where } \eta_{ijk} = \begin{cases} 1, & \left|\hat{\mathcal{Y}}_{ijk} - \mathcal{Y}_{ijk}\right| < \theta \\ 0, & \text{otherwise} \end{cases};$$

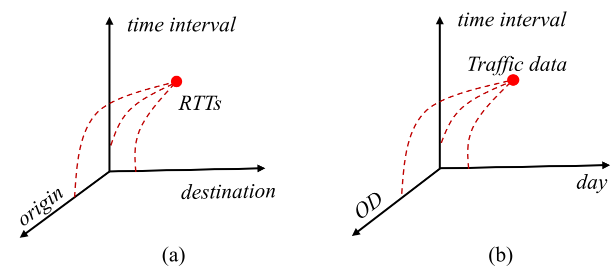

Fig. 4. Illustration of PlanetLab network latency data and Abilene network traffic data as third-order tensors.

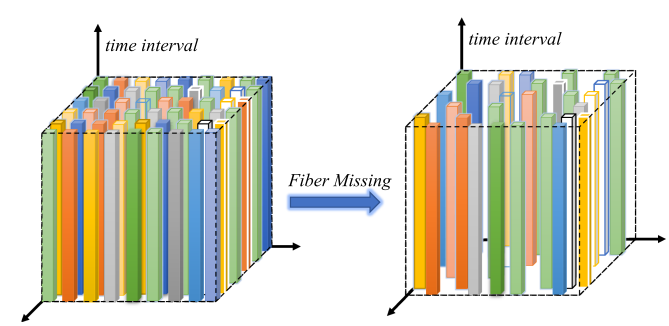

Fig. 5. Illustration of fiber missing scenario.

$$\mathrm{MAE} = \frac{\sum \left|\hat{\mathcal{Y}}_{ijk} - \mathcal{Y}_{ijk}\right|}{n_1 \times n_2 \times n_3}; \quad \mathrm{RRSE} = \frac{\|\hat{\mathcal{Y}} - \mathcal{Y}\|_2}{\|\mathcal{Y}\|_2},$$

where $\theta = 0.05$.

Our proposed algorithm is implemented in Matlab. All experiments are performed on PC with Intel-Core i5-10600KF CPU and 48 GB RAM, and the operating system is Win10.

A. Synthetic Data

In this subsection, we assess performance of our proposed model on synthetic datasets. The ground-truth low-rank tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is generated by two small size factor tensors $\mathcal{A} \in \mathbb{R}^{n_1 \times R \times n_3}$ and $\mathcal{B} \in \mathbb{R}^{n_2 \times R \times n_3}$. In addition, $\mathcal{A}$ and $\mathcal{B}$ obey

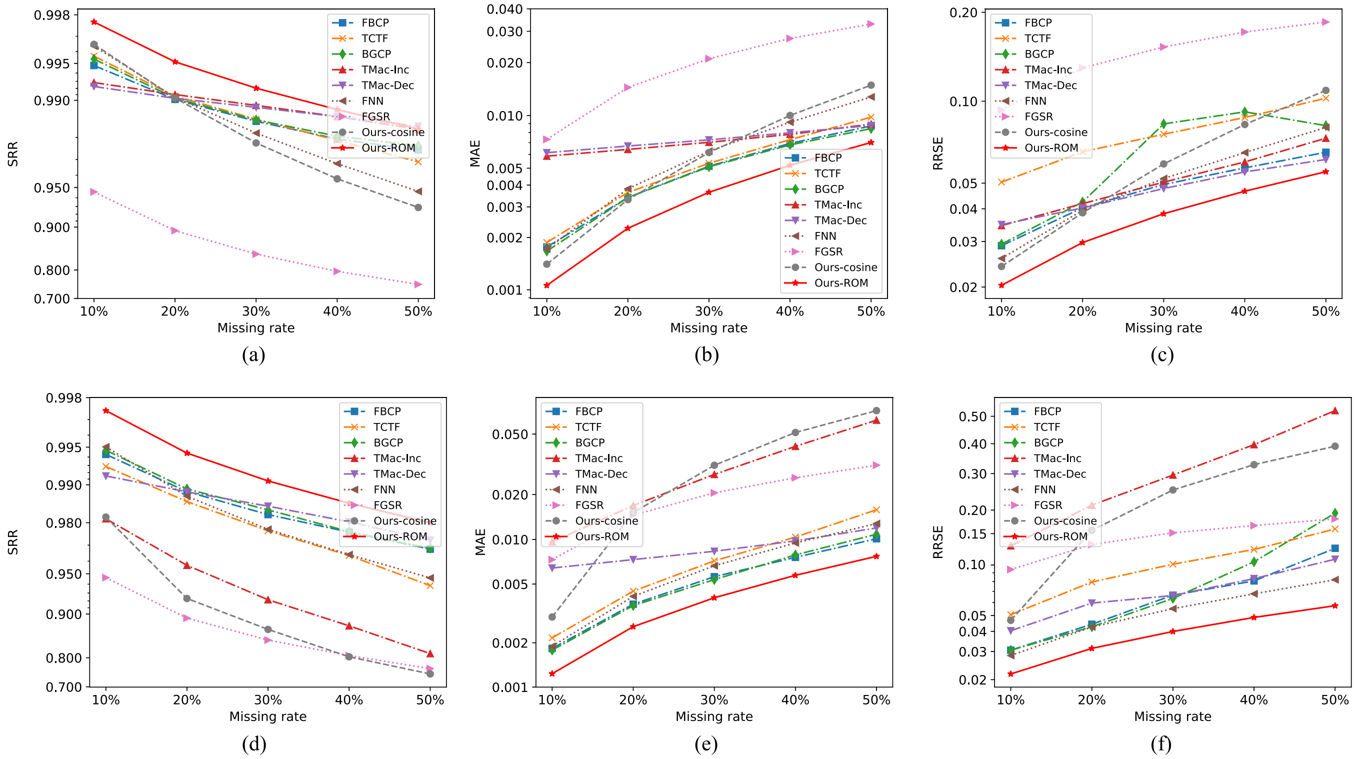

Fig. 6. Performance comparison of our model and other models on *PlanetLab* dataset with missing rates ranging from 10% to 50%. The top row is the performance of SRR, MAE, and RRSE in the random missing scenario, while the bottom row is the performance of SRR, MAE, and RRSE in the fiber missing scenario.

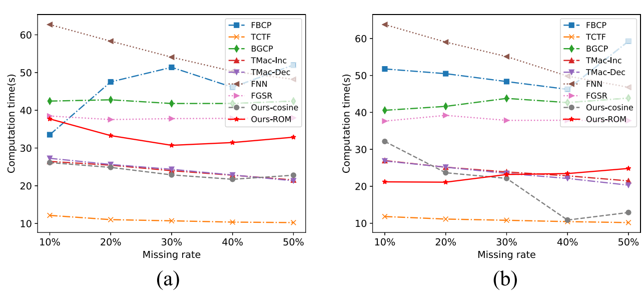

Fig. 7. Comparison of the computation time of our model and other models on the *PlanetLab* dataset with missing rates ranging from 10% to 50%. (a) Random missing scenario. (b) Fiber missing scenario.

Uniform distribution, and low-rank components are constructed by $\mathcal{A} * \mathcal{B}^{\dagger}$. To better reflect the robustness of our model, some noise is added to observed data, the observed tensor is given by $\mathcal{Y} = \mathcal{X} \circ \mathcal{O} + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ with noise parameter $\sigma_\varepsilon^2 = 10^{-2}$. The missing entries obey the Uniform distribution, which is marked by binary tensor $\mathcal{O}$. The synthetic datasets includes tensors whose sizes are $100 \times 100 \times 10$ and $200 \times 200 \times 20$, and we generate different incomplete tensors with missing rate $\rho = [20\%, 40\%, 60\%, 80\%]$.

Table II shows the recovery result of our model (Cosine and ROM based) on the synthetic datasets. It can be seen that the ROM-based model performs better than the Cosine-based model performance on incomplete tensors which have different sizes with or without noise. For the Cosine-based method, the recovery performance is better when the data missing rate is 80% than missing rate with 60%. By analyzing the experimental process, it can be found that under the same circumstances, the result of rank estimation with the missing rate 60% is worse than that with the missing rate 80%, which is the main reason for the poor effect.

Notice that our algorithm can recover most of the data even when the missing rate is 80%. In addition, by comparing the with or without of noise, it can be found that by adding noise $\sigma_\varepsilon^2 = 1E-02$, our algorithm is affected extremely weakly. This shows that our proposed algorithm has strong robustness.

B. Real Network Traffic Measurement Data

1) Datasets Description: We conduct experiments on two real-world network traffic measurement datasets, including a network latency trace dataset (PlanetLab [17]) and network traffic dataset (Abilene[2]).

The PlanetLab dataset consist of round trip times (RTTs) between 490 nodes in the PlanetLab network over 18-time slices, which can be represented as a third-order tensor according to origin-ID, destination-ID, and time-interval. Specifically, as shown in Fig. 4(a), its size is $490 \times 490 \times 18$.

The Abilene network traffic data consists of 12 nodes traffic matrices, with the measured every fifth minute over 168 days, which can be represented as a third-order tensor with a size of $144 \times 288 \times 168$. The details are shown in Fig. 4(b).

[2]http://abilene.internet2.edu/observatory/datacollections.html

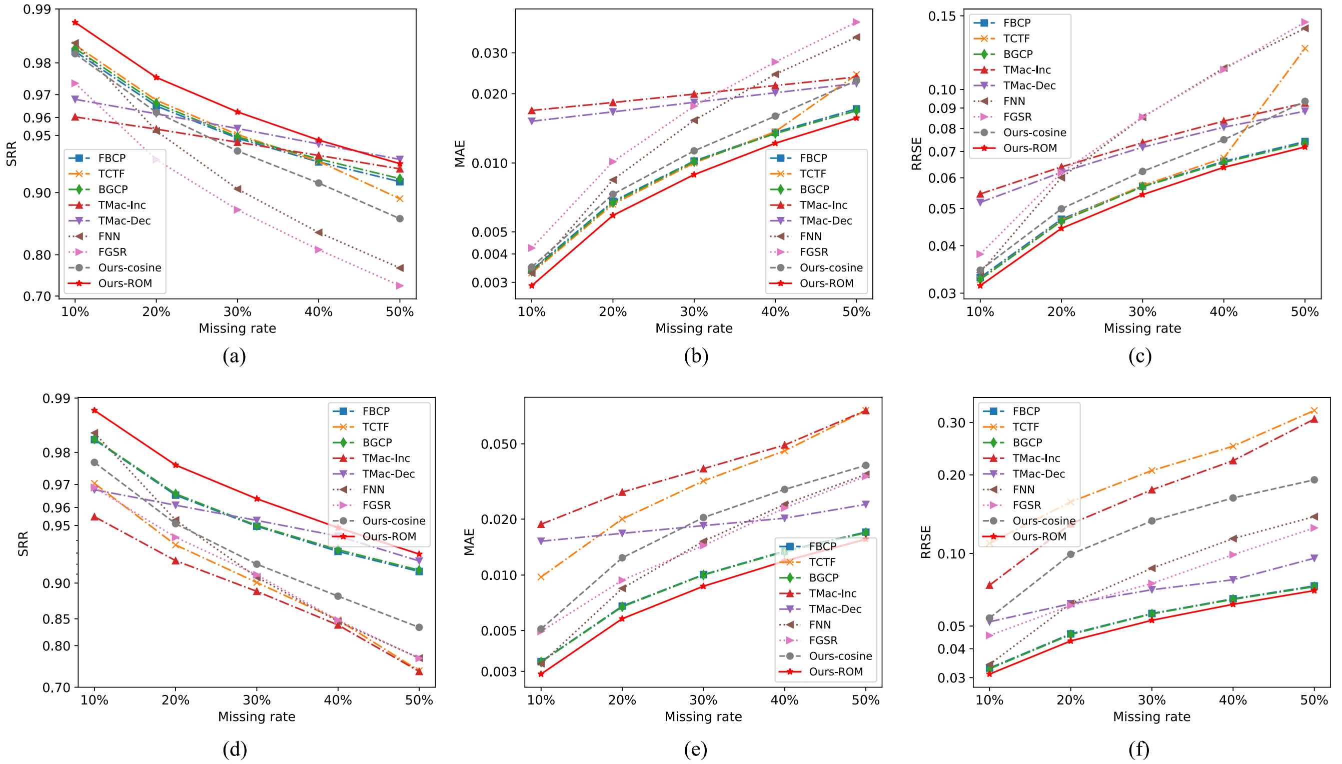

Fig. 8. Performance comparison of our model and other models on *Abilene* dataset with missing rates ranging from 10% to 50%. The top row is the performance of SRR, MAE, and RRSE in the random missing scenario, while the bottom row is the performance of SRR, MAE, and RRSE in the fiber missing scenario.

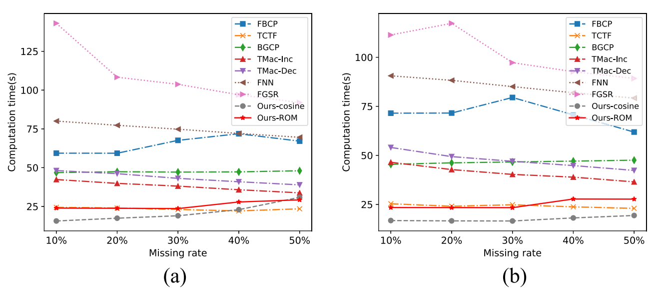

Fig. 9. Comparison of the computation time of our model and other models on the *Abilene* dataset with missing rates ranging from 10% to 50%. (a) Fiber missing scenario. (b) Fiber missing scenario.

2) Random Missing V.s. Fiber Missing: In Section V-A, our proposed model uses random missing for synthetic data. However, in practical applications for recovery of missing network-wide traffic measurement data, data missing can be the result of faults in network communications. Therefore, the missing data in network-wide traffic measurement data is often continuous over time. Considering the above situation, two kinds of data-missing scenarios (random missing and fiber missing) are applied in this subsection. For random missing, $\rho\%$ of the data entries are missed while missing entries obeying normal distribution. For fiber missing, we represent the network traffic data as a third-order tensor, and then $\rho\%$ fibers are missed randomly in the time interval dimension. The details about fiber missing are shown in Fig. 5. In fact, under the same missing ratio, compared with the random missing scenario, the fiber missing will miss more information, and it will be more difficult to recover.

3) Experiments on PlanetLab Dataset: We evaluate our proposed model and other matrices/tensor-based completion models on the PlanetLab network latency trace dataset. Fig. 6 shows the performance of our model and other models under different missing rate. It is clearly shown that our model (ROM-based) obtain the highest performance with the least MAE and RRSE. Fig. 6(a) and (d) compare the successful recovery performance with different missing rates under random missing and fiber missing scenarios. Obviously, our proposed ROM-based transforms model achieves the best results on the evaluation index of SRR.

Fig. 7 shows the computation time of all algorithms under the missing rate from 10% to 50%. Although the algorithm TCTF consumes the least computation time, its performance on network traffic data recovery is ordinary. From a comprehensive point of view, compared with other algorithms, our model (ROM-transform based) has achieved the best performance.

4) Experiments on Abilene Dataset: To better evaluate the recovery performance of our proposed model on the missing network-wide traffic measurement data, we also conduct experiments on the Abilene dataset. Fig. 8 shows the performance of our model and other models under different missing rate scenarios. Obviously, our proposed algorithm outperforms other algorithms in terms of SRR, MAE, and RRSE.

Fig. 9 shows the computation time of all algorithms under the missing rate from 10% to 50% in Abilene dataset. It clearly shows the superiority of our algorithm and TCTF in terms of computation time. However, the performance of the TCTF in recovering missing network traffic data is much lower than our algorithm. Therefore, our model exhibits efficient and accurate performance on the task of missing Abilene network traffic data recovery.

VI. Conclusions and Future Work

Inferring network-wide traffic data from partial network traffic plays an important role in next-generation network systems. At the same time, missing network traffic measurement data is inevitable in network systems. In this article, a transforms-based Bayesian tensor completion method is proposed to recover missing network-wide traffic measurement data. Experiments on synthetic data and two real-world network traffic datasets show that our proposed model outperforms other matrix/tensor-based methods, which can effectively improve network traffic data quality and provide network traffic data assurance for network monitoring, traffic analysis and intrusion detection.

Since the tensor-tensor product with arbitrary invertible linear transforms and tubal rank are defined on a third-order tensor, we only consider recovery 3D data in this article. With the tensor-tensor product with arbitrary invertible linear transforms and tubal rank defining in high-order space, the TBTC can naturally recover high-order data. In addition, the tensor-tensor product and tensor tubal rank depend on given invertible linear transforms, and the different types of data and tasks may depend on appropriate linear transforms. Therefore, it is also an interesting work to explore different linear transforms for different network traffic data analysis tasks, such as network traffic anomaly monitoring, attack detection, *etc*.

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